CHAPTER 9

Combinatorics and counting

S22, S25, S30, S35

Before you begin · Sets and choices

Without a calculator, in six minutes. Answer briefly. Use choice trees and factorials. Then choose the targeted review indicated under the questions.

1. Count the outcomes of two coin tosses.
2. Calculate $4 \times 3 \times 2 \times 1$.
3. Distinguish "choose 2 pupils" from "appoint a president then a vice-president".
4. How many choices of a vowel then a consonant, with 5 vowels and 21 consonants?
5. Write $A \cap \overline{A}$.
6. List the two-element subsets of $\{a, b, c\}$.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Draw the four pairs of results; R31. **2.** Perform multiplications from left to right. **3.** Compare (A,B) and (B,A) according to the students' roles. **4.** Use two levels in the tree; R31. **5.** Look for a element that would satisfy both conditions. **6.** List without changing the order of writing to avoid duplicates. The R questions serve as follow-up material with the teacher, even if their collective routine is planned for later.

Objectives

Choose a counting model; distinguish order and repetition; use binomial coefficients and Pascal's rule.

Course notes

Add or multiply

For finite pairwise disjoint sets A_i, $|\bigcup A_i| = \sum |A_i|$. If the cases overlap, we must correct double counting: $|A \cup B| = |A| + |B| - |A \cap B|$. The number of elements of the product Cartesian $A \times B$ is $|A||B|$. For a sequence of choices, multiply the numbers of possibilities at each stage, provided these numbers are valid for each previous choice considered.

Example. A code of two letters chosen from 26, followed by three digits, with repetition allowed, can take $26^2 \times 10^3$ values. A three-digit number does not start with 0: there are 9×10^2.

Lists and subsets

In a set of n elements, there are n^k ordered lists of length k with repetition; $n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$ lists of k distinct elements, if $0 \leq k \leq n$; $n!$ permutations of all elements. We set $0! = 1$.

Combinations

A combination of k elements among n is a subset of k elements; order does not matter.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}.$$

Why divide by $k!$? Each subset of size k gives exactly $k!$ distinct lists depending on the order of its elements. The number of lists without repetition is therefore $k!\binom{n}{k}$.

Identities and proofs

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Pascal's rule above is used for $0 \leq k < n$; the endpoint formulas can be read directly with the values 1.

Symmetry. Taking the complement sets up a one-to-one correspondence between the subsets of size k and those of size $n-k$.

Pascal, combinatorial proof. In a set of $n+1$ elements, fix one element a. A subset of size $k+1$ contains a (choose k other elements from n) or does not contain it (choose $k+1$ elements among n). The two cases are disjoint and cover every possibility.

Pascal, algebraic proof. For $0 \leq k < n$, put over a common denominator:

$$\frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{n!((k+1)+(n-k))}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n-k)!}.$$

Edge cases are dealt with directly.

Number of subsets. For each of the n elements, choosing "present" or "absent" gives 2^n subsets. Classifying them by size, we count $\sum_{k=0}^{n}\binom{n}{k}$: both counts are equal.

Extension mentioned in the scheme of work. Expanding $(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$ is understood by choosing, in each of the n factors, the term a or b. This formula is not used as a prerequisite for the written tests in this booklet.

Define what is being counted

9.1 Order and repetition: two separate questions

First use: S22.

Worked example · Two letters chosen from A, B, C

With order and repetition, the nine lists are AA, AB, AC, BA, BB, BC, CA, CB, CC. With order without repetition, six lists remain. Without order or repetition, the three objects are $\{A, B\}$, $\{A, C\}$ and $\{B, C\}$. The result changes because the counted object changes, not because we choose another formula.

1st letter / 2nd letter	A	B	C
A	AA	AB	AC
B	BA	BB	BC
C	CA	CB	CC

Cross out the diagonal to forbid repetition; group AB and BA to ignore order.

A1 · Immediate application List the codes of two distinct digits taken from $\{0, 1, 2\}$. How many represent two-digit numbers? Explain the difference between a code and a number.

Property · Addition and product

Disjoint cases are counted by addition. Successive choices are counted by multiplication if each stage offers the stated number of possibilities, given the previous choices. For n distinct objects, the lists of length k without repetition number $n(n-1)\cdots(n-k+1)$; the number of complete permutations is $n!$.

A2 · Partial guidance An outfit has one top from four and one pair of trousers from three. One top and a pair of trousers are red. Count the outfits containing both red items, then exactly one red item.

9.2 Forgetting order without losing possibilities

First use: S30.

Definition · Binomial coefficient

$\binom{n}{k}$ is the number of subsets with k elements of a set with n elements. For $0 \leq k \leq n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. In particular $\binom{n}{0} = \binom{n}{n} = 1$.

Proof. Each subset with k elements gives $k!$ distinct ordered lists. All the subsets give the same number of lists. Dividing the $n!/(n-k)!$ lists without repetition by $k!$ thus counts each subset exactly once. □

Worked example · A team and roles

A team of three students out of eight can be chosen in $\binom{8}{3} = 56$ ways. If one of the three becomes captain, there are $56 \times 3 = 168$ choices. We recover $\binom{8}{2} = 168$ by first choosing the captain. To assign three distinct roles, there would be $8 \times 7 \times 6 = 336$ possibilities.

A3 · Independent choice Ten pupils, six in Terminale and four in Première, form a team of three. Count the teams containing exactly two Terminale students, then at least one student from each level, by two methods.

A4 · Fix a count "Choosing two delegates out of five gives 5 $\times 4 = 20$ possibilities." Explain the mistake if the roles are identical. In which situation would the result 20 be correct?

9.3 Two proofs of Pascal's rule, two readings of a sum

First use: S35 and S37.

Property · Proven identities

For $1 \leq k \leq n$, $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$. In addition, $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

Proof. Single out one element in a set of size $n+1$. The subsets of size k that do not contain it are $\binom{n}{k}$; those which contain it require $k-1$ other elements, i.e. $\binom{n}{k-1}$. The two cases are disjoint and exhaustive. By calculation, put the two fractions over the same denominator $k!(n-k+1)!$ gives a numerator $n![(n-k+1)+k] = (n+1)!$. Finally, a subset of a set with n elements is built by n binary choices (take each element or not), i.e. 2^n subsets; the sorting them by size gives the sum. □

Worked example · A distinguished element

Among five pupils including Lina, a team of two contains Lina (four choices for her partner) or does not contain her ($\binom{4}{2} = 6$ choices). Hence $\binom{5}{2} = 4 + 6 = 10$.

A5 · Immediate application From row 1, 4, 6, 4, 1 of Pascal's triangle, construct the next one and check the sum of its coefficients. Interpret the coefficient 10.

9.4 Complement and double counting

First use: S35.

Worked example · At least one occurrence

A four-digit code in $\{0,1,2,3,4\}$ allows repetition. There are 5^4 codes and 4^4 codes without zero, therefore $5^4 - 4^4 = 369$ codes with at least one zero. The calculation 4×5^3 would count codes containing several zeros several times.

A6 · Choice of strategy Among these codes, count those that contain exactly two zeros. Then count those whose digit sum is odd, without listing the 625 codes.

A first step: For the second question, classify according to the number of positions holding an odd digit: one or three.

One minute, spoken aloud. Explain why systematically dividing a product by a factorial may be wrong: first state what is regarded as the same result.

Delayed review · S41 then S85. Recover the binomial coefficient as a number of paths containing exactly k successes; compare this interpretation with that of a team.

Investigation

Order and repetition in counting - 30 min.

Six different students volunteer to organise a debate.

1. Two pupils are chosen to form a pair without assigning roles. How many pairs can be formed?
2. A president and a secretary are appointed, who must be two different pupils. How many of assignments of the two roles are possible?
3. Three successive presentations are planned. The same pupil may present several times. How many schedules of three presentations are possible?
4. We start again with only four pupils, denoted by A, B, C and D. Write all pairs and all the assignments of the two roles. Then explain why there are 64 schedules of three presentations, without listing them.
5. For each of the three ways of choosing, state whether order matters and whether the same person can be chosen several times.

Progressive exercises

E1 · Direct application. A code is a sequence of four digits chosen from 0, 1, . . . , 9. The repetitions and the digit 0 in first position are allowed. How many codes contain at least one 0?

E2 · Application and justification. Ten athletes run a race with no ties. How many different podiums, with a first, second and third place, can be obtained?

E3 · Method to organise. From seven distinct people, a group of three people is formed containing a person designated in advance. How many groups can be formed?

E4 · Choice of strategy. On a grid, we start from $(0;0)$ to reach $(4;3)$ by performing exactly four moves of one unit to the right and three upwards. How many different paths can be followed?

E5 · Synthesis or proof. A class has eight girls and seven boys. How many groups of four students containing exactly two girls can be formed?

Work from the textbook or a past paper

Barbazo, p. 294, exercise 37. At home after S35: deal with both questions. State whether order matters and whether a card can be chosen more than once before using a binomial coefficient.

Homework

DM9 - Forming a committee - 45 min, after S35.

A team consists of five teachers and seven students, all distinct. A committee is formed of four people.

1. How many different committees can be formed?
2. How many committees contain exactly two pupils?

3. How many committees contain at least one teacher and at least one pupil?

4. From the members of a committee in question 2, we appoint a teacher as chair and a student secretary. Count the possible choices of committee together with these two roles.

5. By counting the committees according to their number of teachers, prove that

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4}.$$

MCQ A · order and repetition

After S25; MCQ B after S35.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. A three-digit code chosen from 0, 1, 2, 3, with repetition allowed, gives:

A. 4×3 B. 3^4

C. 4^3 D. 4!

2. Two distinct roles are given to two people among five. The number of assignments is:

A. 10 B. 20

C. 25 D. 120

3. The number of permutations of four distinct objects is:

A. 16 B. 12

C. 4 D. 24

4. For two disjoint finite sets A, B, $|A \cup B|$ is:

A. $|A| + |B|$ B. $|A||B|$

C. $|A| - |B|$ D. $|A \cap B|$

5. With two cards A and B, the lists of length 2 with repetition are:

A. AB, BA B. AA, BB

C. AA, AB, BA, BB D. A, B

MCQ B · complement and double counting

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. Choosing three pupils among eight with no assigned role is counted by:

A. 8^3 B. $\binom{8}{3}$

C. $8 \times 7 \times 6$ D. 3^8

7. The number of subsets of a five-element set is:

A. 5! B. 10

C. 25 D. 32

8. Pascal's rule gives $\binom{6}{3}$ equal to:

A. $\binom{5}{2}+\binom{5}{3}$ B. $\binom{5}{2}\binom{5}{3}$

C. $2\binom{6}{2}$ D. $\binom{6}{4}+1$

9. An urn contains 3 red and 2 blue distinct balls. The number of pairs without replacement containing at least one red ball is:

A. 6 B. 10

C. 9 D. 3

10. Six people are split into two numbered groups of three. The number of ways is:

A. 10 B. 20

C. 40 D. 6!

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $\binom{n}{0}=0$.

2. Counting with the complement can avoid distinguishing several cases.

3. Drawing three objects simultaneously always gives the objects an order.